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INFORMS JOURNAL ON APPLIED ANALYTICS
Vol. 54, No. 4, July–August 2024, pp. 298–311
ISSN 2644-0865 (print), ISSN 2644-0873 (online)

Improving South Korea's Crystal Ball for Baseball Postseason Clinching and Elimination

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Received: June 17, 2023
Revised: October 7, 2023
Accepted: October 24, 2023
Published Online in Articles in Advance: January 30, 2024

https://doi.org/10.1287/inte.2023.0035

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Abstract. Sports media outlets often report on whether professional teams have clinched a playoff berth or, conversely, have been eliminated from one. Although the U.S. media have reported such results for years, particularly for Major League Baseball, no such equivalent outlets backed by rigorous mathematical models exist in South Korea. Motivated by the hiatus in most international professional sports (with a notable exception being Korean baseball) during the height of the COVID-19 pandemic in March 2020, we derive inspiration for the generation of so-called *magic numbers* by modifying an existing set of integer-programming optimization models that had been developed for U.S. Major League Baseball; the modifications tailor these models to the Korean Baseball Organization League by considering unique rules such as the fact that a game can end in a draw. Our results, which are new to Korean media, have generated interest among Korean baseball fans curious about up-to-date, precise information regarding whether their team still has a chance of qualifying for postseason play or whether it has been eliminated.

Funding: This work was made possible, in part, by a generous donation from Columbia University and supported by award number 2227548 from the National Science Foundation, Division of Civil, Mechanical and Manufacturing Innovation (CMMI).

Keywords: integer-programming applications • Korean baseball • tie-breaking • playoffs

Introduction

The Korean Baseball Organization (KBO) League is the most elite level of baseball played in South Korea and is analogous to the United States' Major Leagues. Founded in 1982, the KBO consists of 10 teams based in nine different cities throughout the country (Figure 1), which are unified into a single league (or division). In Korea, baseball enjoys the most attention of any national sport and is known for its enthusiastic fan base. Since 2015, each team in the KBO plays 144 regular-season games, with each team playing every other team 16 times. Games are allowed to end in a draw. The schedule is a heavy six-game-per-week routine, with Mondays designated for rest (although they can be used to make up games canceled because of inclement weather). The league standings are determined by the teams' winning percentages, which do not include draws. Ties in league standings at the end of the regular season are broken by (simple) rules. Specifically, decreasing by relevance, two-way ties in league standings are won by the team (i) with more wins in the head-to-head series; (ii) that scored more runs in the head-to-head series; and (iii) that finished higher in the standings according to the Korean series (for the top two teams) or in the final standings (for the remaining teams) in the previous year. Two-team ties for first or fifth place are special cases that are broken by a tie-breaking game between the relevant teams. Once the final standings have been determined (i.e., after all ties have been broken), the top five teams compete in a four-round "stepladder playoff system," in which the winner in each round plays the team in the next-highest position (Wikipedia 2022): (i) the wild card game (the fifth-place team versus fourth-place team); (ii) the semi-playoff series (the wild card game winner versus third-place team); (iii) the playoff series (the semiplayoff series winner versus second-place team); and finally (iv) the Korean series (the playoff series winner versus first-place team). Hereafter, we refer to the four-round, stepladder system as the *playoffs*, and we use the term *postseason* to refer to the playoffs plus the tie-breaking games for first and fifth places.

In March 2020, it was announced that KBO was tendering international media rights effective through 2023; the winner of the tender would be able to cover all exhibition, regular-season, All-Star, and postseason games for the elite league (McCullagh 2020). In May

Figure 1. (Color online) Ten Teams Comprising the Korean Baseball Organization's League

Source. Wikipedia (2022).

of the same year, KBO announced that ESPN, jointly owned by the Walt Disney Company and Hearst Communications, had won (Impey 2020). At the time of this writing, the original Korean financial organization, Shinhan Bank, serves as the primary sponsor of the Korean Baseball Organization League (Shinhan Financial Group 2022).

Literature Review

In many countries, sports media outlets have traditionally relied on basic arithmetic to determine whether a team has been eliminated from a playoff spot or, conversely, has clinched a playoff berth. In this case, the correspondingly reported *magic number* reflects the smallest value such that the sum of the number of wins by the first-place team and number of losses by the second guarantees a better season-ending record for the first- versus second-place team (Adler et al. 2002). The magic number, calculated in this basic way, is sufficient, but not necessary, for first-place elimination (Schwartz 1966, Hoffman and Rivlin 1971); Robinson (1991) demonstrates this concept for the 1987 Major League Baseball (MLB) season. For the same sports league, by introducing additional constraints, Adler et al. (2002) formulate integer-linear programming models to determine magic numbers for the following four outcomes: (i) first-place clinch; (ii) first-place elimination; (iii) playoff clinch; and (iv) playoff elimination. The results have appeared for several decades at the time of this writing on the Remote Interactive Optimization Testbed website (riotsports.net), known as Riot (Adler et al. 1996). More recently, Riot has begun to report magic numbers for the National Basketball Association (Husted et al. 2021), the Women's National Basketball Association, and the Korean Baseball Organization. Raack et al. (2014) apply similar models to debunk myths about thresholds in several popular professional sports leagues. For example, they show that, contrary to popular belief, 40 standings points are not necessarily sufficient to avoid relegation in the German Bundesliga.

We focus here on baseball in a country other than the United States, namely South Korea, where a theory postulates that national sports, in general, were established as an underhanded attempt to divert public attention away from the politics of the country at that time, specifically, the rise of a dictator in the early 1970s (Bing 2021). However, given both government and private entrepreneurial support, baseball in Korea has developed into a similar national pastime to American baseball; with a similar draft structure for recruiting and retaining top athletes (Janabel 2020) and with its own characteristics (Figure 2) (ESPN Internet Ventures 2016). For a history of sports in Korea, with attention given to those on the international scene, see Kwak et al. (2017).

Korean baseball garnered increased attention at the start of the COVID-19 pandemic when it was one of only a few team sports being played live; at that point, American baseball analysts ranked Korean baseball fourth in its professional baseball list behind Major League Baseball (United States); Nippon Professional Baseball (Japan); and AAA baseball (United States). The KBO was ranked ahead of AA baseball (United States) in terms of the caliber of its players and its competitiveness on the international scene (Glaser 2020).

Figure 2. (Color online) Jose Bautista Performs a Bat Flip

Source. Jason Swaby (2015), CC-BY-NC-SA 2.0, https://www.flickr.com/photos/jswaby/21990794728.

Some analytical work has been conducted for the Korean Baseball Organization. For example, Bae et al. (2012) project team rankings using statistical methods, whereas Moon et al. (2013) employ Markov chains to predict batters' completions at bat and corresponding runs scored; Ku and Kim (2017) generate indicators to forecast outcomes of particular games. However, these models are predictive rather than prescriptive. Lee (2020) uses an optimization-based technique from graph theory to determine whether a team has been eliminated from the playoffs; the author re-examines an existing method (Ahuja et al. 1993) that solves a max-flow/min-cut model for each team in the league in an arbitrary order and argues that by solving the models in ranked order, teams with worse records than those of already-eliminated teams are, by dominance, also eliminated, thus saving computation time. A fictitious mini-Ivy League and sets of teams from American Major League Baseball serve as test cases. Hockey and soccer (football) are two other examples of major professional sports that allow games to end in draws. Cheng and Steffy (2008) and Ribeiro and Urrutia (2005) present mixed-integer programming models for clinching and elimination in the National Hockey League and Brazilian Championship A Series, respectively. The KBO handles draws in a fundamentally different way than the National Hockey League or any of the major professional soccer leagues. To the best of our knowledge, our models are the first to address the draws as they are accounted for by the KBO.

We extend existing, prescriptive elimination models to determine a suite of magic numbers for the Korean Baseball Organization. Specifically, our contributions are (i) a set of rigorous mathematical programs that determine optimization-based "magic numbers" for each team in the KBO; (ii) methods to efficiently generate solutions to these instances; (iii) the dissemination of the results in the Korean sports media—to our knowledge, these are the most accurate results publicly available; and (iv) the presentation of what is to the best of our knowledge the first examples of clinching and elimination problems whose natural formulations as nonlinear mixed-integer programs can be solved more quickly with leading commercial software than their linearized counterparts.

Traditional Methods

At the time of this writing, popular Korean sports media outlets use traditional magic numbers to establish standings for the KBO League, but even then, this information is disseminated sporadically. One such example occurred in September 2022. On the 22nd of that month, LG had 79 wins, and their magic number for second place was six relative to KT and five relative to Kiwoom. The next day, LG beat Lotte, KT lost to Samsung, and Kiwoom beat Doosan. As a result, LG's magic number for a second-place finish dropped to four against both KT and Kiwoom. As a historical note, this was the first time in 28 years that LG's winning record had been that high and, correspondingly, the associated magic numbers received significant attention from Korean media outlets with reports that this record reduced the magic number needed to confirm LG's second place finish in the league from six to four. It was additionally reported that with two additional wins in the remaining games that season, LG would break its own record for most games won (Hyun-Koo 2022). The link http://eng.koreabaseball.com/Standings/TeamStandings.aspx provides a popular KBO website that lists teams' win-loss-draw records, whereas the site https://mykbostats.com/stats/magic_numbers extends this to traditional magic numbers. However, not only are magic numbers derived through traditional means, but both of these sites are in English, suggesting that most of the traffic is from the United States or other English-speaking countries.

Similarly, the Korean media in the Korean language has been reporting results based on traditional magic numbers. For example, a 2019 article has the headline "Professional baseball SK reduced the winning magic number to 3." Nowhere in the article is any mention made of behind-the-scenes calculations to arrive at this magic number; rather, the article implies that it is derived solely from win-loss-draw records of the various teams in the league (Ilbo 2019). Likewise, a more recent article from 2022, entitled "SSG reduces magic number to 7 with a dominant win over Doosan," reveals no additional calculations (Yunhap News 2022).

Optimization Models

Because it is the number from which others can easily be derived, we focus on the *postseason elimination number*, that is, the minimum number of games a given team k needs to win to remain in contention for postseason play at any given date in the season. We derive this number for each team k at each point during the regular season by running a mixed-integer nonlinear optimization model ($\mathcal{M}_k$) (see Appendix A) every night after a day's worth of games have been played.

The primary inputs for such a model instance are current win-loss-draw records of the KBO teams and the schedule of games remaining in the season. We use statistics such as the number of teams that qualify for postseason play, each team's position in the prior year's standings, and the number of runs (and wins) a team has scored against another to determine tiebreaks between teams in the standings. Primary decision variables are specific to the magic number being

computed but generally consist of the number of additional games a team wins against another team; the number of additional games between two given teams that end in draws; and the total number of wins a team has at the end of the season. From these, auxiliary variable values, such as the total number of wins and losses for a given team, can be computed. In addition, binary variables track logic such as relative win percentages between two teams, the existence of ties between teams, and whether a team wins a tie-break over another team. In the case presented here, the computation of the *postseason elimination number*, the model seeks a set of variable values that determines a minimum-effort (i.e., least number of games that must be won) scenario consisting of a win-loss-draw record for a given team such that it can still qualify for postseason play or show that no such scenario exists.

The minimum-effort *postseason elimination number* is the result of an objective that minimizes the total number of games won by the team in question. That is, the model seeks to determine the fewest possible games for a team in question to win and still qualify for postseason play, subject to 11 sets of constraints that govern the way in which games are played and basic logic: (i) a feasible scenario with respect to the schedule of remaining games (computing wins, losses, and draws accordingly); (ii) a viable and corresponding win percentage (excluding draws); (iii) team ordering relative to its win percentage; (iv) postseason elimination based on relative standings; (v) the computation of draws in games; (vi) indicators regarding whether ties exist in the team standings; (vii) correspondingly, the computation of ties in the standings; (viii) and (ix) two sets of tie-breaking criteria for team standings; (x) the use of tie-breaking criteria (from (viii) and (ix)) in the completed series; and (xi) tie-breaking criteria given the draw count for the team in question, team k, relative to the records (winning percentages) of all other teams. Finally, variables are restricted to nonnegative and/or integer or binary variables, as appropriate. An optimal solution to $(\mathcal{M}_k)$ determines a scenario in which team k qualifies for postseason play with as few wins as possible. If $(\mathcal{M}_k)$ is infeasible, then team k is eliminated from the postseason. Otherwise, team k's postseason elimination number is the value of the objective less the current number of its wins.

The Riot website hosts this and three additional KBO optimization models, providing the following magic numbers for each team: (i) the first-place elimination number, which is the minimum number of games a team must win to avoid being eliminated from first place in the league standings; (ii) the postseason-clinch number, which represents the minimum number of games a team must win to ensure that it finishes the season among the top five teams (possibly in a tie for fifth place); and (iii) the first-place clinch number, which represents the minimum number of games a team must win to ensure that it finishes the season in first place.

Specifically,

- The model for the first-place elimination number is similar to $(\mathcal{M}_k)$ except that the constraints restrict the feasible region to scenarios in which team k finishes in first place.
- The model for the postseason-clinch number is essentially the opposite of $(\mathcal{M}_k)$. That is, it maximizes the number of games that team k wins subject to team k finishing out of contention for the postseason. If the model is infeasible, then team k has clinched a postseason berth. Otherwise, team k's postseason-clinch number is the total number of wins team k has in the scenario less its current number of wins plus one.
- Likewise, the first-place clinch number is determined by solving a model that maximizes the total number of wins team k has in the scenario subject to team k not finishing the season in first place.

These optimization models used to determine magic numbers for the KBO are similar in nature to those solved for Major League Baseball (Adler et al. 2002) and the National Basketball Association (Husted et al. 2021). However, there are significant differences in the technical details because of each league's unique playoff structure and tie-breaking criteria. For example, unlike in the latter professional sports leagues, KBO games may end in a draw, and do so with some regularity. Table 1 illustrates instances in which a team has 10 or more draws at the end of the regular season; in fact, the number of draws exhibits an increased frequency in the past few years.

At the end of the regular KBO season, the teams finishing in first through fifth place are seeded in a playoff tournament to determine the season champion (Wikipedia 2022). If there is a two-way tie for either first or fifth place, the teams involved in the two-way tie play an additional game. In the former case, the team that wins the game is awarded first place in the final regular-season standings, and the loser is relegated to second place. In the latter case, the winning

Table 1. Instances in Which Teams End the Season with 10 or More Draws

Team	Number of draws	Year
Pacific	10	1993
Lotte	11	2004
LG	14	2021
SSG	14	2021
Kia	10	2021
Hanwha	12	2021

Source. https://www.baseball-reference.com/register/league.cgi?id=3ef9029d.

Table 2. KT and Samsung Finished in a Two-Team Tie for First Place at the End of the Regular 144-Game Season

Place	Team	Wins	Losses	Draws	Win percentage
1	KT	76	59	9	0.563
1	Samsung	76	59	9	0.563
3	LG	72	58	15	0.554
4	Doosan	71	65	8	0.522
5	Kiwoom	70	67	7	0.511
6	SSG	66	64	14	0.508
7	NC	67	68	9	0.496
8	Lotte	65	71	8	0.478
9	Kia	58	76	10	0.433
10	Hanwha	49	83	12	0.371

Note. KT won the tie-breaking game and was awarded first place.

team is awarded fifth place (and the lowest seed in the playoffs), and the losing team is relegated to sixth place and does not advance to the playoffs. Table 2 illustrates an example of a two-way tie for first place between KT and Samsung, which occurred at the end of the 2021 season.

For cases in which there is a three-or-more-way tie for fifth place, KBO uses the criteria, decreasing by importance, given in Table 3. These tie-breaking criteria violate necessary conditions for the existence of the first-place elimination threshold for MLB (Adler et al. 2002). For example, let us assume that the team currently in first place has won 50 games and has a first-place-elimination number of zero (i.e., there exists a scenario in which the current first-place team fails to win another game but nevertheless finishes in first place). Let us further assume that the second-place team has already lost the head-to-head series with the first-place team. Because of the first tie-breaking criterion, the team that is currently in second place would have to win at least 51 games to finish in first place. Thus, it is not necessarily true that every team in the KBO has the same threshold for first-place elimination. Table 4 provides all historical ties in KBO team standings at the end of the regular season; one of these is irrelevant (the tie for sixth place in 2011), and, at any rate, the table reveals that there has never been a three-or-more way tie.

Examples

Figure 3 shows the Riot standings for the KBO on the morning of September 19, 2020. The Games Left column shows that Hanwha has 33 games left to play in the season. Hanwha's best possible winning percentage is 0.451 (64 wins, 78 losses, and 2 draws). Kiwoom's worst possible winning percentage is 0.469 (67 wins, 76 losses, and 1 draw). Because Hanwha cannot catch up to Kiwoom, the former team is mathematically eliminated from first place. This is indicated by the word "Elim" in the 1st Place column under the Avoid Elimination From heading. Conversely, the table indicates that NC can clinch first place by winning 32 more games (1st Place column under the Clinch heading). If NC wins 32 of its remaining games and loses the other four, their final winning percentage will be 0.674 (95 wins, 46 losses, and 3 draws). And, if Kiwoom wins all its remaining games, its final winning percentage will be 0.664 (95 wins, 48 losses, and 1 draw). This type of straightforward calculation shows that NC will have the best winning percentage in the KBO if it wins 32 more games.

Our website provides explanations for these and other less obvious examples (https://s2.smu.edu/~olinick/RIOT_Dev-master/KBO_numbers_explanation.html).

By taking the schedule into consideration, we can derive SK's elimination from first place: SK's best possible winning percentage is 0.483; this is better than NC's and Kiwoom's worst possible winning percentages of 0.447 and 0.469, respectively. Therefore, it would appear that there is still at least a mathematical possibility for SK to finish in first place. However, by taking the remaining schedule of games into account, our model can determine that SK cannot finish in first place even if it wins all of its remaining games. For example, suppose that the three remaining games in the series between NC and Kiwoom end in draws and that NC and Kiwoom lose all of their other remaining games. That would give NC a final winning percentage of 0.457 (63 wins, 75 losses, and 6 draws) and Kiwoom a final winning percentage of 0.479 (67 wins, 73 losses, and 4 draws). Because Kiwoom has three games left with KT, KT's worst possible winning percentage in this scenario is 0.455 (65 wins, 78 losses, and 1 draw). If KT loses all of its other remaining games (i.e., all the games against teams other than Kiwoom), then it will lose seven games to LG. Because LG also has seven games left against NC, its worst possible winning percentage in this scenario is 0.539

Table 3. These Tie-Breaking Rules Cover Most Cases in Which There Are Three-or-More-Way Ties in the Final KBO Standings (Kim 2022), Ordered Decreasing by Importance, and in Which a Tie Is Broken by the First Rule in the Sequence for Which There Is a Discrepancy in Outcome Between the Two Teams

Criterion	Description of tie-break
1	Team with more wins against other teams in the tie
2	Team that scored more runs against other teams in the tie
3	Team that finished higher in the previous year's final standings

Table 4. Instances in Which the Standings Revealed Two-Way Ties at the End of the Regular Season; There Has Never Been a Three-or-More-Way Tie

Teams	Positions of tied teams	Year
Kia–Hanwha	3–4	1994
Hanwha–LG	6–7	2011
Doosan–SK	1–2	2019
Doonsan–LG	3–4	2020
KT–Samsung	1–2	2021
Kiwoom–KT	3–4	2022

(76 wins, 65 losses, and 3 draws), which is better than SK's best possible winning percentage of 0.483. This is only one of many possible scenarios, but it turns out that in every possible scenario, some team other than SK finishes in first place.

Let us consider another example: LG's first-place clinch number of 32 cannot be verified without considering the remaining match-ups during the season. By winning all 32 of their remaining games, LG can achieve a winning percentage of 0.667 (94 wins, 47 losses, and 3 draws), which is less than NC's and KT's maxima of 0.702 and 0.671, respectively. However, if LG wins all of its remaining games, then it will win seven against NC and another seven against KT, reducing their maximum winning percentages to 0.652 and 0.622, respectively.

Finally, the asterisk (*) in the standings table indicates that Kiwoom does not have a first place clinch number because it cannot guarantee a first place finish even by winning all of its remaining games. As noted, Kiwoom and NC only have three games left in their season series. That means that even if Kiwoom wins all of its remaining games to achieve a 0.671 winning percentage, NC can still finish in first place with a winning percentage as high as 0.681 (i.e., 96 wins, 45 losses, and 3 draws). Although Kiwoom does not have a first place clinch number, it is still possible for the team to finish in first place. The Riot standings indicate that Kiwoom must win at least two more games (69 total) to avoid elimination from first place. Similarly, KIA does not have a first place clinch number but can still finish in first place as long it wins at least 10 more games (69 total). In KIA's case, the scenario involves KIA winning tie-breakers against other teams with the same winning percentage.

Data and Results

We use data for the parameter values from a popular baseball website hosted by the Korean Baseball Organization (http://eng.koreabaseball.com/), which provides background on player performance, such as pitching and batting, in the league. More relevant for our purposes are lists of all the games a team has completed and has yet to play. For the former, the winning team and the runs scored by both teams are available and are constantly changing as the season progresses. These data populate our optimization models.

In addition to the playoff elimination model ($\mathcal{M}_k$) detailed in Appendix A, we solve models for the first-place elimination and first-place and postseason clinch numbers for each team (see Appendix C). We implement these mixed-integer nonlinear programs in

Figure 3. (Color online) Riot Standings for the KBO on the Morning of September 19, 2020

							Additional Wins Needed To			
							Clinch		Avoid Elimination From	
Team	W	L	D	GB	PCT	Games Left	1st Place	Postseason	1st Place	Postseason
NC	63	42	3	0.0	0.600	36	32	23	6	0
Kiwoom	67	48	1	1.0	0.583	28	*	20	2	0
KT	62	47	1	3.0	0.569	34	*	25	7	0
LG	62	47	3	3.0	0.569	32	32	24	7	0
KIA	59	49	0	5.5	0.546	36	*	29	10	0
Doosan	58	49	4	6.0	0.542	33	*	27	11	0
Lotte	56	51	1	8.0	0.523	36	*	31	13	2
Samsung	49	60	2	16.0	0.450	33	*	*	20	9
SK	38	74	1	28.5	0.339	31	*	*	Elim	20
Hanwha	31	78	2	34.0	0.284	33	*	*	Elim	27

the AMPL modeling language, Version 20221013 (AMPL 2009), and solve them in parallel with up to 16 threads using the Gurobi Solver, Version 10.0.0 (Gurobi Optimizer 2010), on an HP DL380 with 768 GB of memory and Dual 20-core Intel Xeon processors running at 2.20 GHz under the CentOS Linux (release 7.9) operating system. We give Gurobi a 20-minute time limit, which is sufficient in most cases. If Gurobi terminates with a solution that is not provably optimal, we solve the model a second time using the incumbent as an initial solution, and the dual bound (rounded up or down, as appropriate, to the nearest integer value) from the first solution as a constraint on the value of the objective function; we change Gurobi's search strategy to emphasize proving optimality. If this solve fails to prove optimality within the same time limit, then we terminate with a "timed out" model and report the best objective function value found up to that point.

Table 5 provides a representative sample of solution times over the course of the 2022 KBO season. The *Games Played* column indicates the percentage of the 720 games in the regular season that had been played by the given date. The next two columns give the number of problem instances that timed out on the given date and the largest gap measured in terms of games (i.e., the absolute gap). The *Total CPU Time* column yields the total time required to solve the 40 mixed-integer nonlinear programs in hours, minutes, and seconds (i.e., in `hh:mm:ss` format). Early in the season, it requires several hours to calculate the magic numbers, and there are multiple postseason clinch and/or elimination numbers that are potentially under- or over-represented by one game because of an optimality gap. However, the solution times decrease rapidly after the first third of the season and are negligible during the final third of the season when the numbers are most interesting to fans. For example, the largest instance from April 11, 2022, early in the season, possesses 1,499 rows, 797 columns, and 4,564 nonzero values in the constraint matrix; of the 1,499 rows (constraints), 20 contain bilinear terms, and of the 797 columns (variables), 30 are continuous and 767 are integer (with 312 of these being binary). By contrast, the largest instance from October 10, 2022 (i.e., later in the season), possesses 163 rows (with 20 of them containing bilinear terms), 124 columns (30 continuous variables and 94 integers of which 55 are binary), and 501 nonzeros in the constraint matrix.

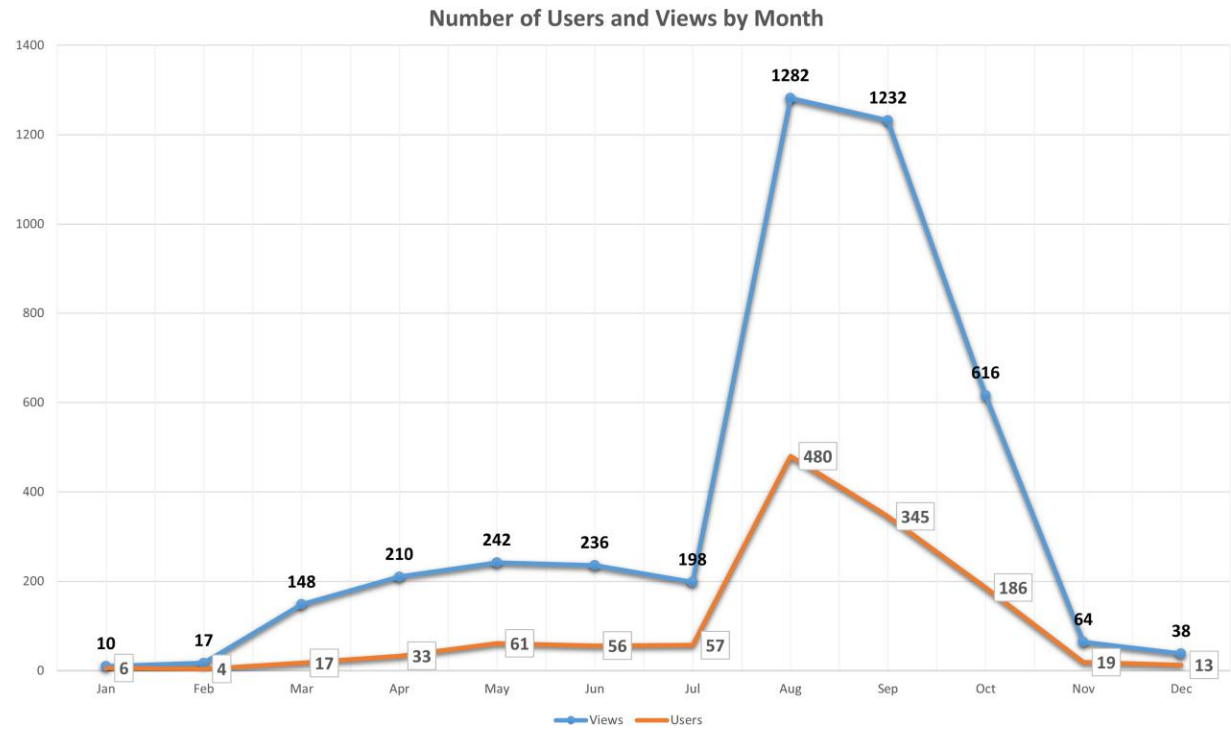
Riot's website reports only on win-loss-draw records and corresponding magic numbers for each sport it hosts; by contrast, it does not report on other newsworthy items such as updates on player health, analysis of various games, and/or player trades. It is therefore difficult to generate traffic to the website early in the season because the magic numbers are not particularly informative; specifically, insufficient games have been played to yield any conclusions other than the facts that no team has been eliminated from first place or the playoffs, nor has any team definitively clinched first place or a playoff spot. Because other websites offer more information, viewers interested in the standings table are drawn elsewhere. As the magic number outcomes become interesting, we see a rise in viewership. Figure 4 shows, for the three-year period starting in January 2020 and ending in December 2022, the number of views and number of users; we aggregate both statistics across all three years by month. Viewership is highest in the fall just before the end of the regular season in mid-October.

Korean news reporters have begun to focus on the magic numbers Riot has generated, as opposed to

Table 5. Approximate Percent of Games Played, Number of Time-Outs Across Both Solves (if Applicable), Total Run Time (Measured in Hours:Minutes:Seconds), Maximum Absolute Gaps, and Total Run Time Considering Each Given Team k Based on Selected Dates During the 2022 KBO Season

Date	Games played	Time-outs	Maximum absolute gap	Total CPU time
04/11/22	5.56%	3	1	4:08:47
04/25/22	13.61%	5	1	4:55:45
05/09/22	21.94%	6	1	4:38:12
05/23/22	30.14%	3	1	2:22:09
06/06/22	37.36%	3	1	2:32:27
06/20/22	45.28%	0	0	0:30:38
07/04/22	52.5%	1	1	1:46:10
07/18/22	58.06%	0	0	0:40:10
08/01/22	63.75%	0	0	0:01:54
08/15/22	70.56%	0	0	0:00:34
08/29/22	78.06%	0	0	0:00:10
09/12/22	85.42%	0	0	0:00:04
09/26/22	93.33%	0	0	0:00:01
10/10/22	97.64%	0	0	0:00:01

Figure 4. (Color online) Riot Website Viewership for the Kbo Page for a Three-Year Time Horizon over Which the Website Has Reported Magic Numbers



traditional magic numbers, in some of their media outlets. For example, sports writer Bae Ji-heon interviewed Sam Kim for an article that appeared on the Korean sports website spochoo.com in 2020 (Bae 2020); Figure 5 shows an excerpt from that article. The text in bold above the standings table translates to "NC, with 10 more wins, secures first place … With 4 more wins, [it will] advance to [the postseason]." The caption explains that the magic numbers in the Riot standings table indicate that NC needs just 10 more wins to secure first place and just four more wins to secure a postseason berth; furthermore, there are scenarios in which NC can make the postseason—even in first place—without winning another game.

A quote from a subsequent article featuring Riot (Bae 2022) is shown in Figure 6. The first paragraph above the table explains the magic numbers, and the line just above the table states that the four teams in positions 1 through 4 have almost clinched postseason berths; furthermore, the next three teams, KIA, Doosan, and NC, can still hope to finish in fifth place on their own (i.e., without relying on teams they do not play for the remainder of the season to lose games). In the article, we learn that SSG can still clinch first place, even if it fails to

Figure 5. (Color online) Riot Kbo Standings Featured on Korean Sports Website Spochoo.com on October 10, 2020

NC, 10승만 더 올리면 1위 자력 확정...4승 더하면 PS 진출 확정

							클린치 넘버		탈락 방지 넘버	
팀	승	패	무	승차	승률	잔여경기	1위	5강	1위	5강
NC	77	46	4	0	0.626	17	10	4	0	0
KT	72	55	1	7	0.567	16	*	10	2	0
LG	71	56	3	8	0.559	14	*	10	4	0
키움	74	59	1	8	0.556	10	*	8	0	0
두산	69	56	4	9	0.552	15	*	12	5	0
KIA	66	60	0	12.5	0.524	18	*	17	8	1
롯데	64	61	1	14	0.512	18	*	18	11	3
삼성	56	70	3	22.5	0.444	15	*	*	Elim	13
SK	43	86	1	37	0.333	14	*	*	Elim	Elim
한화	42	85	2	37	0.331	15	*	*	Elim	Elim

10월 9일까지 경기 기준 리오트 순위표. NC는 10승만 더 하면 1위 자력 확정, 4승만 더 하면 5강이 확정된다. 또 남은 경기에서 1승도 거두지 못해도 1위로 시즌을 마칠 가능성이 있고, 5강 역시 마찬가지다(통계출처=리오트 연구팀 홈페이지)

Figure 6. (Color online) Riot Kbo Standings Featured on Korean Sports Website Spochoo.com on August 12, 2022

RIOT 순위표의 '1위 클린치 넘버'는 해당 팀이 남은 시즌 1위를 자력 확정하는 데 필요한 승수를, '포스트시즌 클린치 넘버'는 해당 팀이 포스트시즌에 자력 진출하는 데 필요한 승수를 보여준다. 또 탈락 넘버는 1위 경쟁, 가을야구 경쟁에서 탈락하지 않기 위해 반드시 이겨야 하는 최소 경기 수를 보여주는 숫자다.

SSG부터 KT까지 1~4위는 가을야구 거의 확정, 두산-NC는 아직 자력 5강 희망 있다

							클린치		탈락넘버	
팀명	승	패	무	게임차	승률	잔여경기	리그우승	포스트시즌	리그우승	포스트시즌
SSG	69	30	3	0	0.697	42	35	19	0	0
LG	59	38	1	9	0.608	46	*	30	10	0
키움	59	42	2	11	0.584	41	*	29	10	0
KT	52	45	2	16	0.536	45	*	36	16	0
KIA	49	49	1	19.5	0.500	45	*	40	20	3
두산	43	53	2	24.5	0.448	46	*	45	25	9
NC	42	53	3	25	0.442	46	*	45	27	10
롯데	43	55	4	25.5	0.439	42	*	*	26	9
삼성	41	57	2	27.5	0.418	44	*	*	28	11
한화	31	66	2	37	0.320	45	*	*	38	21

win any of its remaining games. The synopsis at the beginning of the article (translated from Korean) reads:

- The KBO League outcomes are based on the RIOT ranking table developed by U.S. professors and their students.
- SSG's first place clinch is virtually confirmed … Even if it loses the rest of its games, it can finish in first place, depending on the other teams' records.
- Even if Lotte and Samsung win the rest of their games, they will not necessarily be in the top five.
- There is a significant possibility of KT finishing in fourth place, and Doosan and NC still have a chance for postseason play.

After making a few assessments about whether a team can clinch a playoff spot, the article continues: "The RIOT Leaderboards provide clinch numbers and elimination numbers that are improvements over the traditional magic numbers. It is explained that when a team with a tie in the ranking occurs, it has a comparative advantage over the existing magic number by reflecting the characteristics of the KBO League, such as the KBO's ranking criteria and drawn games." The site then shares a screen capture of RIOT's numbers as of August 12, 2022, and analyzes them. The article concludes with "The RIOT research team discloses the KBO League leaderboard every day on its website (https://s2.smu.edu/~olinick/RIOT_Dev-master/KBO_standings.html). Major League Baseball, National Basketball Association, and Women's National Basketball Association leaderboards are also released." Sam Kim is quoted in an article that mentions RIOT in *Chosun* (Jang 2020), one of the leading newspapers in South Korea; RIOT magic numbers were cited in a story about the Samsung Lions' prospects for the postseason in 2021 (Kim 2021). In this way, Korean baseball fans are being made aware of improvements over traditional magic numbers for assessing whether their teams have clinched a playoff spot or have been eliminated from postseason play. Furthermore, they are being drawn to the RIOT website, which is available in both Korean and English.

Conclusions

Although the RIOT website has generated magic numbers for Major League Baseball for a quarter of a century at the time of this writing, no attempts had been made to disseminate the equivalent numbers for international baseball leagues; correspondingly, neither had international media outlets provided this information. By using models of clinch and elimination numbers for American baseball as a template that we modify accordingly, we generate this information for the Korean Baseball Organization, which has received attention in the U.S. media, specifically, from ESPN, owing to its flamboyant style (e.g., the "bat flipping"), persistence of play during the COVID-19 pandemic, and its international roster. Somewhat counterintuitively, the mixed-integer *nonlinear* formulations solve more quickly than their linearized counterparts with the state-of-the-art solver we use. Future efforts entail continuing to increase traffic on the website, socializing models for the Women's National Basketball Association in the United States, and expanding the suite of models to other types of international teams.

Acknowledgments

The authors thank Nelson "Trey" Winbush for help in data collection, Brayden Park for marketing efforts, Reginald "RJ" Phillips for help in conducting computational tests, and Eojin Han for help with Korean-to-English translation and general feedback on the paper. The authors acknowledge the inputs of two anonymous referees and an editor whose comments helped to strengthen the paper.

Appendix A. Mathematical Formulation for Postseason Elimination

The following formulation is an integer program for a given team k to determine the number of wins needed to preclude its elimination from the postseason. If the model is infeasible, there is no scenario in which team k can qualify for the postseason, that is, team k has been eliminated from the postseason. This integer program, $(\mathcal{M}_k)$, is run for each team k at the end of each day of the season (i.e., after all games have been played) to determine if such a scenario exists given the current standings. There are 11 sets of constraints and domain restrictions that govern the *postseason elimination* problem for a given team k. Variable values indicate the outcome for a particular scenario that presents, for the team in question, a path toward a playoff spot. If no such path exists, that is, if the constraints given the data render the model infeasible, then the team in question has been eliminated. We detail each set of constraints as follows and provide specific explanations in each set. Table A.1 provides the notation.

Table A.1. Notation

Sets	
$\mathcal{F}$	Team pairs between which no more games remain in the season
$\mathcal{R}$	Criteria used to break ties, ordered decreasing by position in the hierarchy
$\mathcal{T}$	All teams in the league
$\mathcal{T}_k = \mathcal{T} \setminus \{k\}$	All teams in the league other than k
Dynamic sets	
$\mathcal{I}_k$	Teams with the same winning percentage as team k, including team k itself, in the scenario
Parameters	
$g_{tt'}$	Number of games left in the season between team t and team t'
$\hat{\imath}_t$	Current number of draws for team t
$\bar{\imath}$	Maximum number of draws for any team in a scenario
k	Team in question
ℓ_t	Current number of losses for team t
M	Sufficiently large number
m	Sufficiently small number
n	Number of teams that qualify for postseason play
p_t	Team t's position in last year's final standings
$\hat{r}_{tt'}$	Current number of runs team t has scored against team t'
$\hat{w}_t$	Current number of wins for team t
$w_{tt'}$	Current number of wins team t has against team t'
Variables	
Continuous variables	
N_t	Current win percentage of team t, expressed in decimal form
Integer variables	
$A_{tt'}$	Number of additional runs team t scores against team t' in the scenario
G_t	Total number of games team t wins against the other teams in $\mathcal{I}_k$, if team t is tied with team k; zero if team t is not tied with team k
I_t	Total number of games team t draws in the scenario
L_t	Total number of games team t loses in the scenario
R_t	Total number of runs team t scores against the other teams in $\mathcal{I}_k$ in the scenario, if team t is tied with team k; zero if team t is not tied with team k
W_t	Total number of games team t wins in the scenario
$X_{tt'}$	Number of additional games team t wins against team t' in the scenario
$Y_{tt'}$	Number of additional times team t and t' draw in the scenario
Binary variables	
$\alpha_{tt'}$	1 if team t wins the tie-break over team t' in the scenario, 0 otherwise
$\beta_{tt'}$	1 if team t has a higher win percentage than team t' in the scenario, 0 otherwise
$\omega_{tt'}$	1 if team t finishes higher in the standings than team t' in the scenario after any relevant tie-breaking criteria have been applied, 0 otherwise
$\hat{I}^0$	1 if team k is tied with no other teams in the scenario standings, 0 otherwise
$\hat{I}^1$	1 if team k is tied in the scenario standings with exactly one other team, 0 otherwise
$T_{tt'r}$	1 if team t wins the tie-break over t' by criterion r in the scenario, 0 otherwise
τ_{tj}	1 if and only if team t has total of j draws in the scenario, 0 otherwise
Z_t	1 if there is a tie between team t and team k in the scenario, 0 otherwise
$\hat{Z}_{tt'}$	1 if teams t and t' are both tied with team k in the scenario, 0 otherwise

Problem $(\mathcal{M}_k)$

A.1. Objective Function

Minimize

$$W_k. \tag{A.1}$$

Objective Function (A.1) minimizes the number of wins needed for team k to preclude its elimination from the postseason. If the model is infeasible, there is no scenario in which team k can qualify, and, correspondingly, team k has been eliminated from the postseason.

subject to:

A.2. Games and Results

$$X_{tt'} + X_{t't} + Y_{tt'} = g_{tt'} \qquad \forall t, t' \in \mathcal{T} : t \neq t' \tag{A.2a}$$

$$W_t = \hat{w}_t + \sum_{t' \in \mathcal{T} : t \neq t'} X_{tt'} \qquad \forall t \in \mathcal{T} \tag{A.2b}$$

$$L_t = \ell_t + \sum_{t' \in \mathcal{T} : t \neq t'} X_{t't} \qquad \forall t \in \mathcal{T} \tag{A.2c}$$

$$I_t = \hat{\imath}_t + \sum_{t' \in \mathcal{T} : t \neq t'} Y_{tt'} \qquad \forall t \in \mathcal{T} \tag{A.2d}$$

$$Y_{tt'} = Y_{t't} \qquad \forall t, t' \in \mathcal{T} : t \neq t' \tag{A.2e}$$

Constraint (A.2a) ensures that each of the remaining games between team t and team t' is either won by one of the two teams or ends in a draw. Constraint (A.2b) calculates the total number of wins for team t as the number of games team t has already won and the remaining number of games team t wins. Constraint (A.2c) calculates the total number of losses for team t as the total number of games team t has already lost and the remaining number of games team t' wins against team t. Constraint (A.2d) calculates the total number of games team t draws as the sum of the current number of games team

t has already drawn and the remaining number of games team t draws with team t'. It is unnecessary, but convenient, to use both $Y_{tt'}$ and $Y_{t't}$ in the formulation. Constraint (A.2e) ensures consistency in these two variable values.

A.3. Win Percentage

$$N_t = \frac{W_t}{W_t + L_t} \quad \forall t \in \mathcal{T} \tag{A.3}$$

Using Constraints (A.2b) and (A.2c), we compute the win percentage in Constraint (A.3) as the quotient of the total number of games won and the sum of games won and games lost. The denominator is a variable because games can end in draws, which are not included in this computation. Therefore, (A.3) is nonlinear. It has an exact linearization, but computation with the linearized constraint runs more slowly. However, we clear the denominator before optimizing.

A.4. Drawn Games

$$I_t = \sum_{j=0}^{\max\{\bar{\imath},\hat{\imath}_t\}} j \cdot \tau_{tj} \qquad t \in \mathcal{T} \tag{A.4a}$$

$$\sum_{j=0}^{\max\{\bar{\imath},\hat{\imath}_t\}} \tau_{tj} = 1 \qquad t \in \mathcal{T} \tag{A.4b}$$

$$I_t \le \max\{\bar{\imath},\hat{\imath}_t\} \qquad t \in \mathcal{T} \tag{A.4c}$$

Constraint (A.4a) counts draws for each team, Constraint (A.4b) ensures that each team has a nonnegative number of draws, and Constraint (A.4c) limits the number of draws a team has to its current total or to a default maximum. We use a value of 14 for $\bar{\imath}$ given the historical numbers in Table 1. Computational gains can be made from setting $\bar{\imath}$ to a lower number earlier in the season and increasing it throughout.

A.5. Tie Indicators in the Standings

$$\hat{I}^0 + \hat{I}^1 \le 1 \tag{A.5a}$$

$$2 \cdot (1 - \hat{I}^0) - \hat{I}^1 \le \sum_{t \in \mathcal{T}_k} Z_t \tag{A.5b}$$

$$\sum_{t \in \mathcal{T}_k} Z_t \le |\mathcal{T}_k| - (|\mathcal{T}_k| - 1) \cdot \hat{I}^1 \tag{A.5c}$$

$$\sum_{t \in \mathcal{T}_k} Z_t \le |\mathcal{T}_k| \cdot (1 - \hat{I}^0) \tag{A.5d}$$

Constraint (A.5a) ensures that one or both of $\hat{I}^0$ and $\hat{I}^1$ are equal to zero. When $\hat{I}^0 = 1$, the left-hand side of (A.5b) is zero, and the right-hand sides of (A.5c) and (A.5d) are $|\mathcal{T}_k|$ and zero, respectively. Thus, $\hat{I}^0 = 1$ forces $\sum_{t \in \mathcal{T}_k} Z_t = 0$ as required (i.e., no teams are tied with team k). When $\hat{I}^1 = 1$, the left-hand side of (A.5b) is one, and the right-hand sides of (A.5c) and (A.5d) are one and $|\mathcal{T}_k|$, respectively. Thus, $\hat{I}^1 = 1$ forces team k to be tied in the standings with exactly one other team. When $\hat{I}^0 = \hat{I}^1 = 0$, the left-hand side of (A.5b) is two, and the right-hand sides of (A.5c) and (A.5d) are both $|\mathcal{T}_k|$. Thus, $\hat{I}^0 = \hat{I}^1 = 0$ forces team k to be tied with at least two (and at most $|\mathcal{T}_k|$) other teams. This allows for multiway ties.

A.6. Ties in the Standings

$$\alpha_{tk} + \alpha_{kt} = Z_t \qquad \forall t \in \mathcal{T}_k \tag{A.6a}$$

$$T_{ktr} + T_{ktr} \le 1 \qquad t \in \mathcal{T}_k, r \in \mathcal{R} \tag{A.6b}$$

$$1 - Z_t \ge \max\{N_t - N_k, N_k - N_t\} \qquad t \in \mathcal{T}_k \tag{A.6c}$$

$$m - (N_t - N_k) \le Z_t + (1+m) \cdot \beta_{kt} \qquad t \in \mathcal{T}_k \tag{A.6d}$$

$$m - (N_k - N_t) \le Z_t + (1+m) \cdot \beta_{tk} \qquad t \in \mathcal{T}_k \tag{A.6e}$$

$$m - (N_k - N_t) \le Z_t + (1+m) \cdot (1 - \beta_{kt}) \qquad t \in \mathcal{T}_k \tag{A.6f}$$

$$m - (N_t - N_k) \le Z_t + (1+m) \cdot (1 - \beta_{tk}) \qquad t \in \mathcal{T}_k \tag{A.6g}$$

$$Z_t + \beta_{kt} + \beta_{tk} = 1 \qquad t \in \mathcal{T}_k \tag{A.6h}$$

$$Z_t + Z_{t'} \le \hat{Z}_{tt'} + 1 \qquad t, t' \in \mathcal{T}_k : t \ne t' \tag{A.6i}$$

$$2\hat{Z}_{tt'} + (\beta_{kt} + \beta_{tk}) + (\beta_{kt'} + \beta_{t'k}) \le 2 \qquad t, t' \in \mathcal{T}_k : t \ne t' \tag{A.6j}$$

These constraints determine the relative order of team $t \in \mathcal{T}_k$ and team k in the standings. Constraint (A.6a) guarantees that team t or team k wins the tie-breaker if they are tied. Constraint (A.6b) ensures that at most one team can win tie-breaking criterion r. Constraints (A.6c)–(A.6g) ensure that the variable Z_t, representing whether there is a tie between teams t and k, is one if there is a tie and zero otherwise by measuring the difference between the teams' winning percentages. Constraint (A.6h) ensures that only one of the three variables $\in \{\beta_{tk}, \beta_{tk}$, or $Z_t\}$ equals one; that is, either team t has a strictly larger winning percentage than team k, or vice versa, or team t and team k are tied, respectively. In the latter case, tie-break criteria 1 and/or 2 are valid. Constraints (A.6i) and (A.6j) articulate that if team t is tied with team k and t' is tied with k, then the variable denoting that both teams are tied with team k is activated.

A.7. Tie-Break Criterion 1

$$G_t = \sum_{t' \in \mathcal{I}_k \setminus \{t\}} (w_{tt'} + X_{tt'}) \quad \forall t \in \mathcal{T} \tag{A.7a}$$

$$1 - (1+M) \cdot (1 - T_{tt'1}) \le G_t - G_{t'} \le M \cdot T_{tt'1} \quad \forall t, t' \in \mathcal{T} : t' \in \mathcal{I}_k \setminus \{t\} \tag{A.7b}$$

Set $\mathcal{I}_k$ comprises team k and all teams with the same winning percentage as team k. Constraint (A.7a) calculates the number of games that teams in set $\mathcal{I}_k$ win against the other teams in the set. For a pair of teams t and t' in set $\mathcal{I}_k$, Constraint (A.7b) determines whether team t or team t' wins the tie-break by having more wins against teams in the set.

A.8. Tie-Break Criterion 2

$$R_t = \sum_{t' \in \mathcal{I}_k \setminus \{t\}} (\hat{r}_{tt'} + A_{tt'}) \quad \forall t \in \mathcal{T} \tag{A.8a}$$

$$1 - (1+M) \cdot (1 - T_{tt'2}) \le R_t - R_{t'} \le M \cdot T_{tt'2} \quad \forall t, t' \in \mathcal{T} : t' \in \mathcal{I}_k \setminus \{t\} \tag{A.8b}$$

Constraint (A.8a) calculates the number of runs that teams in set $\mathcal{I}_k$ score against each other. Constraint (A.8b) determines whether team t or team t' has more runs against all teams in the set of tied teams; the team with more runs wins this tie-break criterion.

Tie-Breaking for Completed Series

$$T_{tt'1} = 1 \quad \forall t, t' \in \mathcal{F} \ni w_{tt'} > w_{t't}, t \ne t' \tag{A.9a}$$

$$T_{t't1} + T_{tt'1} = 0 \quad \forall t, t' \in \mathcal{F} \ni w_{tt'} = w_{t't}, t \ne t' \tag{A.9b}$$

$$T_{tt'2} = 1 \quad \forall t, t' \in \mathcal{F} \ni \hat{r}_{tt'} > \hat{r}_{t't}, t \ne t' \tag{A.9c}$$

Constraints (A.9a)–(A.9c) establish which, if any, of the two tie-breaking criteria are valid for pairs of teams that have already completed their season series. These constraints only apply if there are no more games left between teams t and t', in which case we can set the tie-breaking variables to fixed values a priori to expedite solutions.

A.9. Criteria Hierarchy

$$2^{|\mathcal{R}|+1}\cdot(1-Z_t)+\sum_{r\in\mathcal{R}}2^{|\mathcal{R}|-r}\cdot T_{tkr}+2^{|\mathcal{R}|}\cdot\alpha_{kt}\geq\sum_{r\in\mathcal{R}}2^{|\mathcal{R}|-r}\cdot T_{ktr}\ \ \forall t\in\mathcal{T}_k \tag{A.10a}$$

$$2^{|\mathcal{R}|+1}\cdot(1-Z_t)+\sum_{r\in\mathcal{R}}2^{|\mathcal{R}|-r}\cdot T_{ktr}+2^{|\mathcal{R}|}\cdot\alpha_{tk}\geq\sum_{r\in\mathcal{R}}2^{|\mathcal{R}|-r}\cdot T_{tkr}\ \ \forall t\in\mathcal{T}_k \tag{A.10b}$$

$$T_{tt'3}=1\ \ \forall t,t'\in\mathcal{T}\ni p_t<p_{t'},t\neq t' \tag{A.10c}$$

Constraints (A.10a)–(A.10c) indicate which team wins the tie-break and qualifies for postseason play by arranging the tie-break criteria in order of execution. If criterion 1 is satisfied for either team t or team t', then that team wins the tie-break and qualifies for postseason play. If criterion 1 is not satisfied, then criterion 2 is examined for whether team t or team t' breaks the tie and enters postseason play (Table 3).

A.10. Team Ordering

$$\omega_{tt'}+\omega_{t't}=1\ \ \forall t,t'\in\mathcal{T}:t\neq t' \tag{A.11}$$

Constraint (A.11) ensures that either team t or team t' is ranked higher in the standings after any relevant tie-breaking criteria are applied, either by playing an additional game or according to the rules given in Table 3.

A.11. Postseason Elimination

$$\sum_{t\in\mathcal{T}_k}\beta_{tk}\leq n-1 \tag{A.12a}$$

$$N_k+\omega_{tk}\geq N_t+\alpha_{tk}\ \ \forall t\in\mathcal{T}_k \tag{A.12b}$$

$$\sum_{t\in\mathcal{T}_k}\omega_{tk}\leq n-1+\hat{I}^1 \tag{A.12c}$$

Constraint (A.12a) ensures that at most $n-1$ teams (one fewer than the number of teams that qualify for postseason play) have a strictly better winning percentage than team k. Constraint (A.12b) establishes whether team t or team k finishes higher in the standings based on win percentage and, if necessary, the tie-breaking criteria. Correspondingly, Constraint (A.12c) limits the number of teams that finish higher in the standings than team k. Specifically, when $\hat{I}^1=1$, team k is tied with exactly one other team. In this case, Constraint (A.12c) allows up to n teams to be ranked higher in the final standings than team k, permitting a scenario in which team k is tied for nth place with exactly one other team, regardless of whether that team is ahead of team k based on the tie-breaking criteria. That is, team k may finish in position $n+1$ according to the tie-breaking criteria as long as it is in a two-way tie for nth place based on win percentage. Such a scenario is illustrated in Table 2: KT (team k) is guaranteed a postseason game with Hanwha to determine which team goes to the playoffs. Thus, $\hat{I}^1=1$ allows for scenarios in which team k finishes the season in a two-way tie for nth place (or better), regardless of the tie-breaking criteria.

When $\hat{I}^1=0$, Constraint (A.12c) is tightened so that no more than $n-1$ teams finish higher in the final standings than team k. That is, this condition illustrates the scenario in which team k places nth or higher, based on win percentage and, if necessary, the tie-breaking criteria listed in Table 3. Table 2 illustrates a scenario in which KT is team k; in each case, $\omega_{tk}=1$ for Samsung and $\omega_{tk}=0$ for all other teams. In fact, that year, KT won the tie-break and the championship.

A.12. Nonnegativity and Integrality

$$0\leq N_t\leq 1 \qquad \forall t\in\mathcal{T} \tag{A.13a}$$

$$W_t,I_t,L_t,G_t,R_t\geq 0\ \text{ integer} \qquad \forall t\in\mathcal{T} \tag{A.13b}$$

$$X_{tt'},Y_{tt'},A_{tt'}\geq 0\ \text{ integer} \qquad \forall t,t'\in\mathcal{T} \tag{A.13c}$$

$$Z_t\ \text{ binary} \qquad \forall t\in\mathcal{T} \tag{A.13d}$$

$$\hat{Z}_{tt'},\alpha_{tt'},\omega_{tt'},\beta_{tt'}\ \text{ binary} \qquad \forall t,t'\in\mathcal{T} \tag{A.13e}$$

$$\tau_{tj}\ \text{ binary} \qquad \forall j\in\{0,\ldots,\max\{\bar{\imath},\hat{\imath}_t\}\},t\in\mathcal{T} \tag{A.13f}$$

$$T_{tt'r}\ \text{ binary} \qquad \forall t,t'\in\mathcal{T},r\in\mathcal{R} \tag{A.13g}$$

$$\hat{I}^0,\hat{I}^1\ \text{ binary} \tag{A.13h}$$

Finally, Constraints (A.13a)–(A.13c) ensure that the appropriate variables assume continuous or integer nonnegative values. Constraints (A.13e)–(A.13h) enforce binary restrictions.

As a final implementation note, we observe that for the tie-breaking constraints (both criteria 1 and 2), $\mathcal{I}_k$ is a dynamic set that is not known a priori; we use it for notational convenience but cannot code the model this way. Instead, we substitute Constraints (A.14a)–(A.14d) for Constraint (A.7a).

$$G_k=\sum_{t\in\mathcal{T}_k}(w_{kt}+X_{kt})Z_t \tag{A.14a}$$

$$G_t=(w_{tk}+X_{tk})Z_t+\sum_{t'\in\mathcal{T}_k\setminus\{t\}}(w_{tt'}+X_{tt'})\hat{Z}_{tt'}\ \ \forall t\in\mathcal{T}_k \tag{A.14b}$$

$$G_k\leq\sum_{t\in\mathcal{T}_k}(w_{kt}+g_{kt}) \tag{A.14c}$$

$$G_t\leq MZ_t\ \ \forall t\in\mathcal{T}_k \tag{A.14d}$$

Because Constraint (A.8a) also uses the set $\mathcal{I}_k$, it is replaced with constraints similar to (A.14a)–(A.14d) in the code. In what follows, we discuss numerical values for big M and little m, and we then detail how to modify the model to determine the other magic numbers, namely first-place elimination, first-place clinch, and postseason clinch.

Appendix B. Defining Big *M* and Little *m*

The values of big M and little m are tailored to the constraints in which they are used. For example, in Constraints (A.6d)–(A.6g), the number of games played is 144, so the quotient expressed in Table B.1 captures the smallest point differential as the reciprocal of the product of the total number of games played and that number less one. In general, if n games are played, then this differential can be expressed as $1/(n-1)-1/n=1/(n(n-1))$. The value for big M in Constraint (A.7b) is the sum of the current wins for team t and the number of games remaining between team t and all other teams less the number of wins that team t' has already accrued. In a similar manner, the value for big M in Constraint (A.8b) equals the minimum of 300 (the maximum theoretically possible based on historical values) and the

Table B.1. We Provide the Values for Sufficiently Large Values of Big M and Sufficiently Small Values for m and the Corresponding Rationale

Constraint number	Value of big M or little m	Rationale
(A.6d)–(A.6g)	$\frac{1}{144\times143}$	The (absolute value of the) smallest possible difference between unequal winning percentages
(A.7b)	$\hat{w}_t + \sum_{t''} g_{tt''} - \hat{w}_{t'}$	Team t wins all remaining games and team t' wins no other games
(A.8b)	$\min\{300, \hat{r}_{tt'} - \hat{r}_{t't} + \bar{r} \cdot \sum_{t''} g_{tt''}\}$	Existing difference in runs plus maximum possible remaining for team t (where $\bar{r}$ is based on historical run differentials)
(A.14d)	$\hat{w}_t + \sum_{t'} g_{tt'}$	Team t wins all of its remaining games

sum of the number of runs team t has against team t' less the number of runs team t' has against t plus a theoretical maximum that is taken to be the product of the maximum number of runs ever scored in a game ($\bar{r}$, historically 27, so we use a conservative value of 30) and the total number of games team t has yet to play against any other team. The value in Constraint (A.14d) mimics that given for Constraint (A.7b) but disregards the number of games that team t' has won.

Appendix C. Mathematical Formulations for First-Place Elimination and First-Place and Postseason Clinches

This appendix provides the modifications to the playoff elimination model ($\mathcal{M}_k$) required to obtain the other three magic numbers.

C.1. Elimination from First Place

The mixed-integer nonlinear program for determining whether team k has been eliminated from first place is derived from the playoff-elimination problem ($\mathcal{M}_k$) by replacing Constraint (A.12a) with the following constraints to ensure that team k is at least tied for first place in any feasible scenario.

$$\sum_{t\in\mathcal{T}_k} \beta_{tk} = 0 \tag{C.1a}$$

$$\sum_{t\in\mathcal{T}_k} \omega_{tk} \le \hat{I}^1 \tag{C.1b}$$

If this mixed-integer nonlinear program has a feasible solution, then we cannot eliminate team k from first place, and the corresponding first-place elimination number is calculated in the same way in which the playoff elimination number is computed.

C.2. Clinching First Place

The mixed-integer nonlinear program for determining team k's first-place clinch number is derived from problem ($\mathcal{M}_k$) as follows:

1. W_k is maximized rather than minimized.
2. Constraint (A.12a) is dropped.
3. Constraint (A.12b) is replaced by

$$N_t + \omega_{kt} \ge N_k + \alpha_{kt} \quad \forall t \in \mathcal{T}_k. \tag{C.2}$$

4. Constraint (A.12c) is replaced by

$$\sum_{t\in\mathcal{T}_k} \omega_{kt} \le |T_k| - 1 + \hat{I}^1. \tag{C.3}$$

The combination of (C.2) and (C.3) prevents scenarios in which team k clinches first place. If $N_k > N_t$ for every $t \in \mathcal{T}_k$, then $\hat{I}^1 = 0$ because team k has a strictly better winning percentage in the scenario than every other team. This, in turn, would imply that $\omega_{kt} = 0$ for at least one team t to satisfy (C.3). However, this would lead to a contradiction because (C.2) cannot hold with $N_k > N_t$ and $\omega_{kt} = 0$. When $\hat{I}^1 = 0$, it is possible that team k is tied for first place with two other teams, in which case (C.2) and the team-ordering constraints, (A.11), ensure that at least one of the tied teams wins the tie-breaker over team k. If $\hat{I}^1 = 1$, then team k is tied with exactly one other team. If it is a tie for first place, then neither team has clinched because the two teams must play a postseason game to determine first and second place. We run instances of this model to determine the first-place clinch number, if applicable. If the corresponding instance is feasible, the resulting objective function value represents the maximum number of games team k wins without finishing in first place. If the model is infeasible, then team k has already clinched first place. In the former case, we report $(W_k - \hat{w}_k) + 1$ as team k's first-place clinch number.

C.3. Clinching a Postseason Berth

The postseason clinch model determines, for a given team k, the number of wins needed to clinch a postseason berth. The model maximizes the number of games team k can win, W_k, and yet fail to qualify for the playoffs. The mixed-integer nonlinear program for determining team k's postseason clinch number is derived from the mixed-integer nonlinear program for determining team k's first-place clinch number by replacing (C.3) with

$$\sum_{t\in\mathcal{T}_k} \omega_{kt} \le (|\mathcal{T}_k| - n) + n\hat{I}^1, \tag{C.4}$$

$$\sum_{t\in\mathcal{T}_k} \beta_{kt} \le (|\mathcal{T}_k| - n - 1) + (n+1)(1 - \hat{I}^1). \tag{C.5}$$

If $\hat{I}^1 = 0$, then the tie-breaking rules listed in Table 3 apply to the scenario. In this case, Constraint (C.4) ensures that team k is not positioned sufficiently high in the standings to earn a postseason berth, and Constraint (C.5) is automatically satisfied. Conversely, the tie-breaking rules listed in Table 3 do not apply to the scenario if $\hat{I}^1 = 1$. In this case, Constraint (C.4) is automatically satisfied, and Constraint (C.5) ensures that team k is not positioned high enough in the standings to earn a postseason berth by forcing at least n other teams to have better winning percentages. Thus, team k misses the playoffs in the scenario either by finishing tied for the last playoff spot and losing the tie-breaker or by finishing behind at least n other teams. If the model is feasible, then Riot reports that team k's postseason clinch number is one more than the number of games it wins in the scenario, $(W_k - \hat{w}) + 1$. If the model is infeasible, then there is no scenario in which team k does not qualify for the postseason; that is, team k has clinched a postseason berth.

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Verification Letter

RIOT sports numbers have appeared in the Korean news media, e.g., in the article "NC needs 10 wins to become the winner, and Lotte needs to win every game to advance to the top 5," reported in Spochoo.com by Bae Ji-heon on October 10, 2020. The article begins: "American professor's research team 'Riot' has created a new calculation method to replace the magic number, making headlines." The full article can be found in Korean here: http://www.spochoo.com/news/articleView.html?idxno=76446.

A complete verification letter is on file.

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